Even the supermassive black holes at the centers of galaxies, with masses millions or billions of times that of the Sun, are destined to eventually fade and disappear.

Ordinary matter—the stuff making up stars and planets and gas and dust—suffers a similar, if less dramatic fate.

Most particles of matter are known to be, at some level, unstable. If left alone long enough, they decay into other things, dropping in mass and energy in the process. A neutron, for example, will eventually decay into a proton, an electron, and an antineutrino. While we've never seen a proton decay experimentally, we have reason to believe that can happen too, if you're willing to wait something like 10^{33} years. At that point even hydrogen atoms, which have been persisting as the most numerous atoms in the universe since the Big Bang itself, will finally cease to be.

The distant future of a universe governed by dark energy in the form of a cosmological constant is one of darkness, isolation, emptiness, and decay. But this slow fade is just the beginning of the ultimate end: the Heat Death.

The name "Heat Death" might sound like a misnomer for a state of the cosmos that is colder and darker than anything else in the history of creation. But in this case, the term "heat" is a technical physics term, not meaning "warmth" but rather "disordered motion of particles or energy." And it's not the death *of* heat, but a death *by* heat. It's the disorder in particular that kills us. Which is why we need to take a moment to talk about entropy.

Entropy is perhaps one of the most essential, versatile, and tragically obscure topics in all of science. It shows up everywhere—not just in the physics of everything from balloons to black holes, but also computer science, statistics, and even economics and neuroscience. Entropy is usually explained in terms of disorder. The more disordered a system, the higher its entropy. A pile of puzzle pieces has higher entropy than a completed puzzle; a scrambled egg has higher entropy than an intact one. In cases where "disorder" is not an immediately obvious property, you can think of entropy as a measure of how free or unconstrained the elements of the system are. To be concrete, a completed puzzle has low entropy because there's only one way for all the pieces to be arranged to make the puzzle whole, whereas a pile of pieces can be in any of a number of configurations and still successfully constitute a pile.